Walks,Paths,Distances, Connected Components, Friend of a Friend

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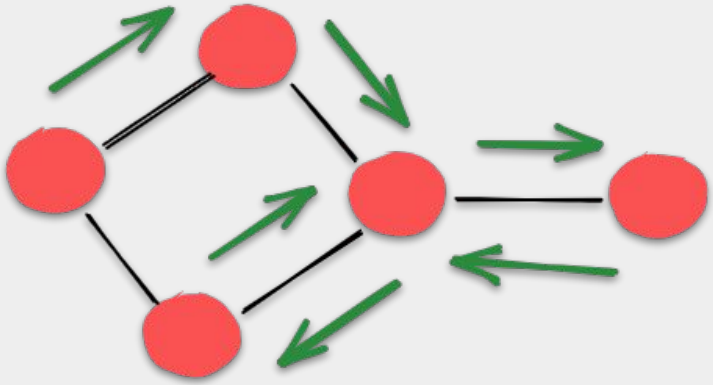
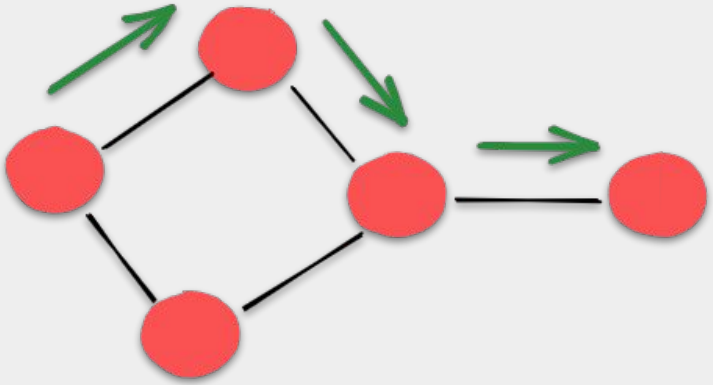
A network really shines when you use it for what it is for: **exploring its connections** .

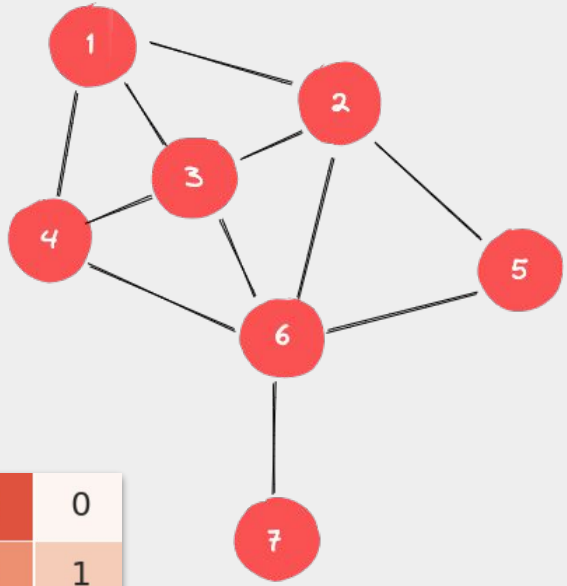


An example of a walk of lenght six in the network



An example of a path of lenght three in the network

Walks and Matrices

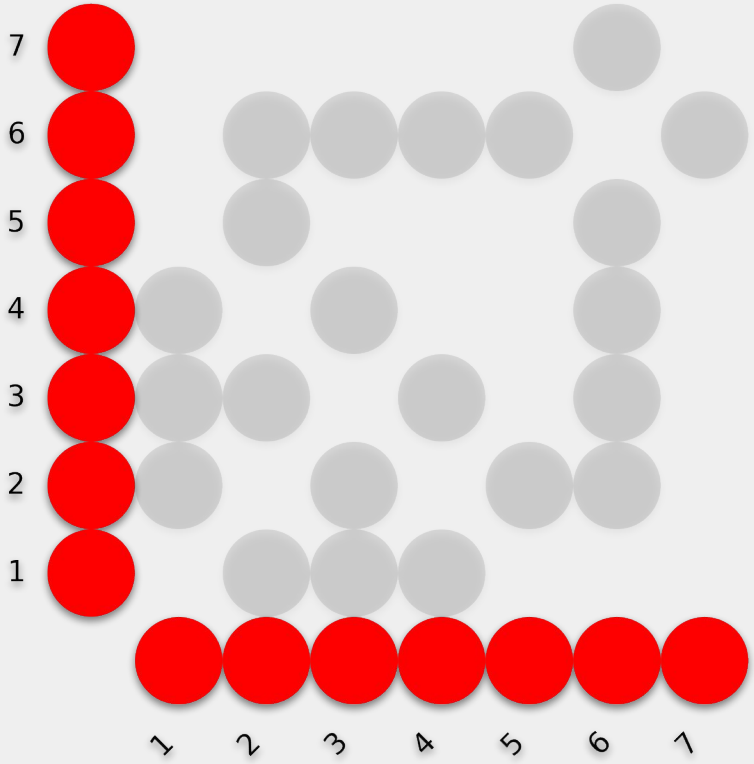


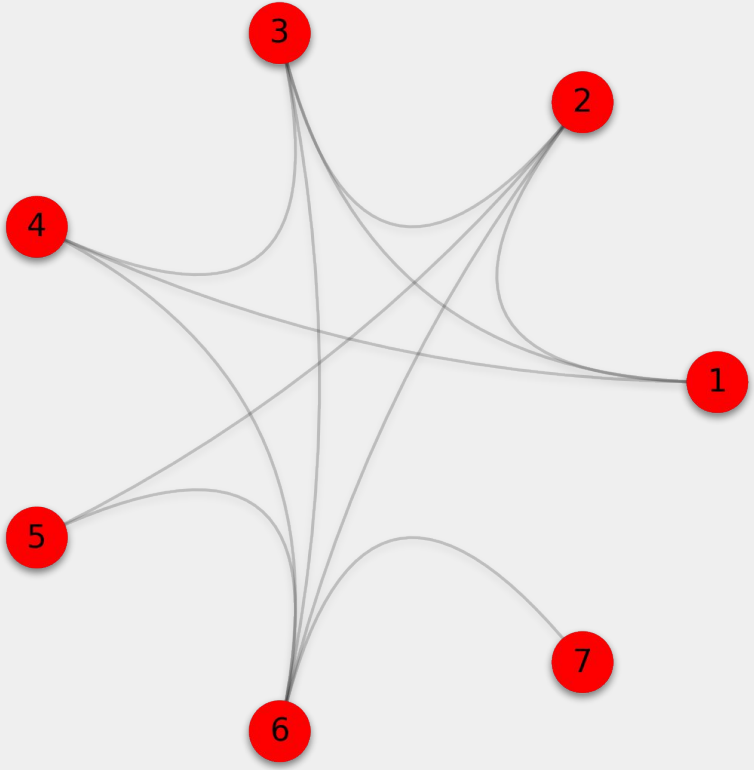
Adjacent Matrix (A)

0	1	1	1	0	0	0
1	0	1	0	1	1	0
1	1	0	1	0	1	0
1	0	1	0	0	1	0
0	1	0	0	0	1	0
0	1	1	1	1	0	1
0	0	0	0	0	1	0

A^2

3	1	2	1	1	3	0
1	4	2	3	1	2	1
2	2	4	2	2	2	1
1	3	2	3	1	1	1
1	1	2	1	2	1	1
3	2	2	1	1	5	0
0	1	1	1	1	0	1

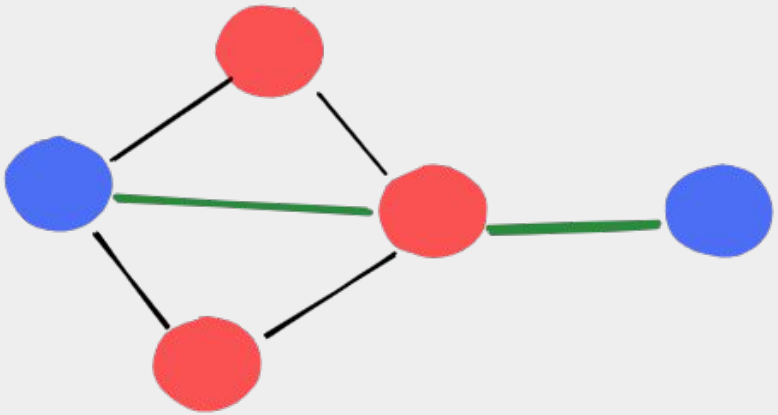




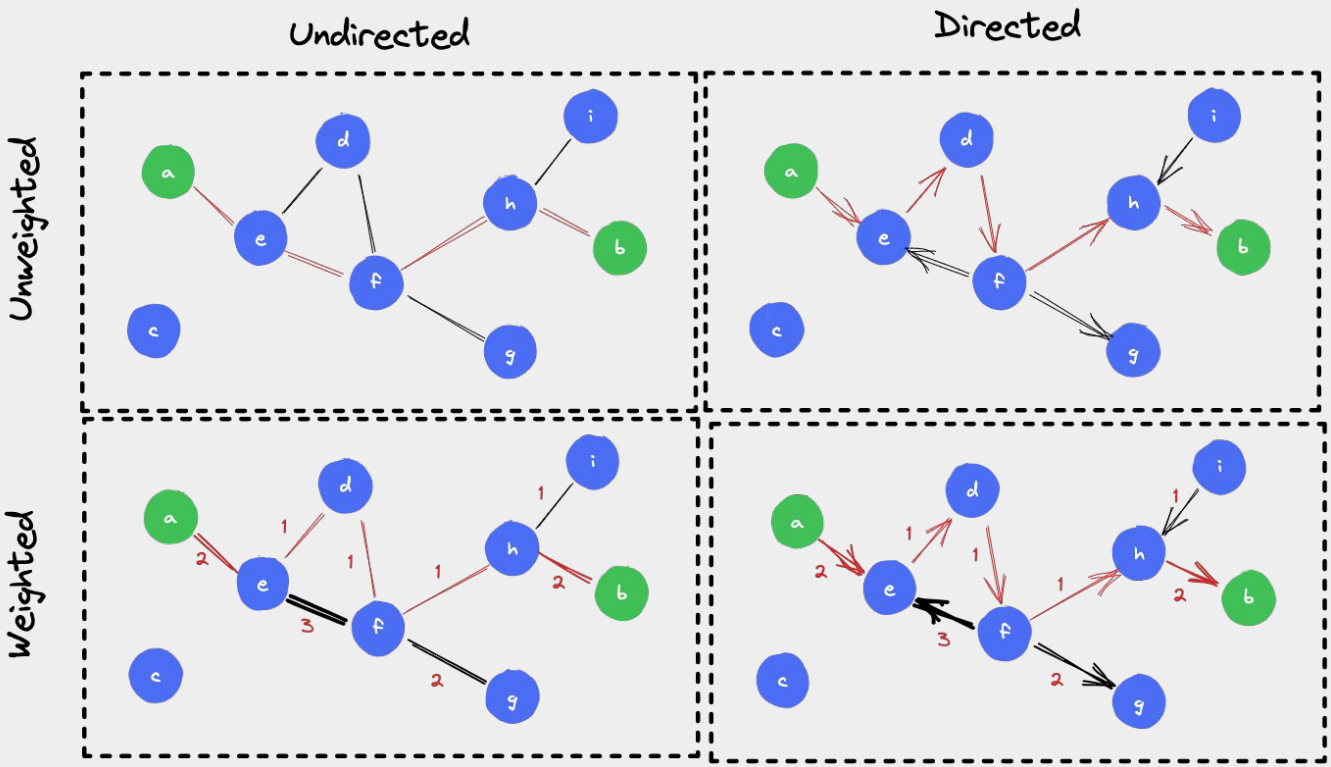
https://ericmjl.github.io/nxviz

Distance

The concept of a **path** is the basis of the definition of **distance** among nodes in a network. The natural distance measure between two nodes is defined as the minimum number of links that must be traversed in a path connecting the two nodes. Such a path is called the **shortest path**, and its length is called the shortest-path length.



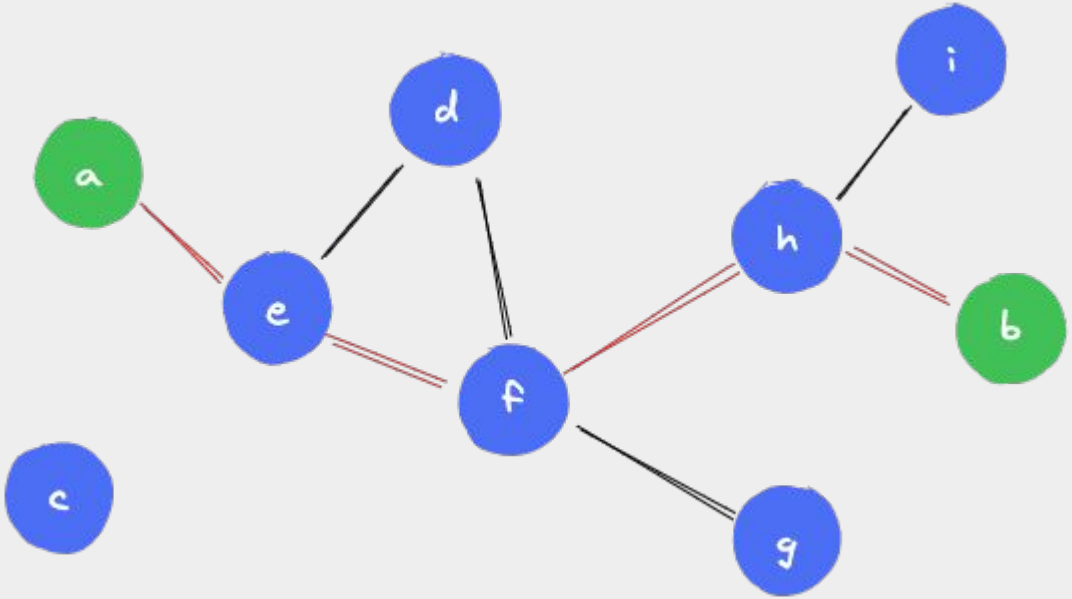
Shortest Path



Average Shortest Path Length

$$\langle l \rangle = \frac{\sum_{ij} l_{ij}}{\binom{N}{2}} = \frac{2\sum_{ij} l_{ij}}{N(N-1)}$$

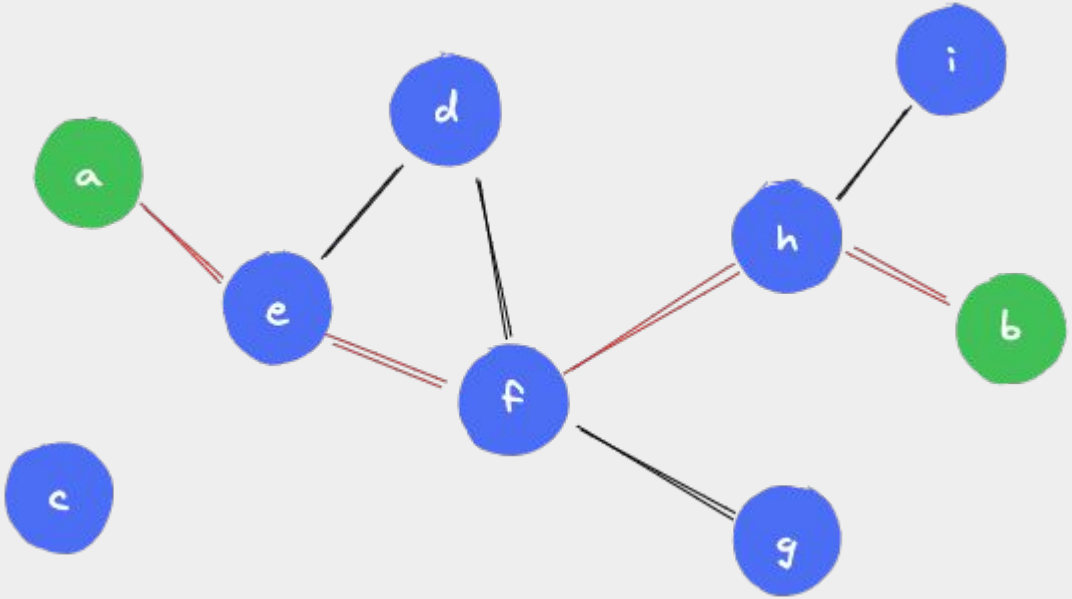
$$\langle l \rangle = \frac{\left(\sum_{ij} \frac{1}{l_{ij}}\right)^{-1}}{\binom{N}{2}}$$



Diameter of Network

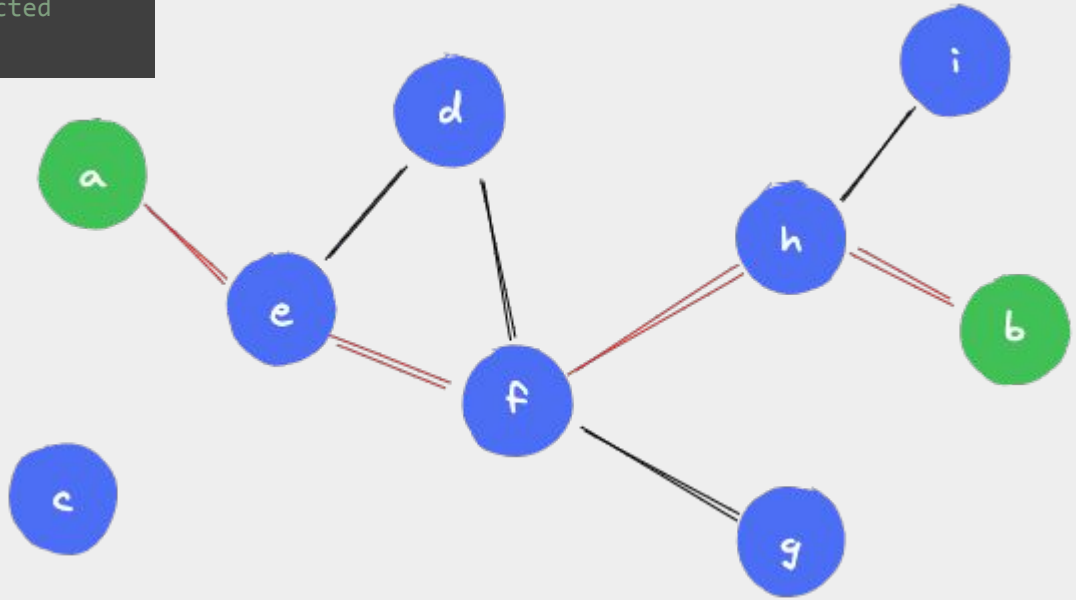
$$l_{max} = \max_{ij} \; l_{ij}$$

(src, dest)	(b,c)	....
(a,b)	-	
a - e - f - h - b	(b,d)	
(a,c)	b - h - f - d	
-	(b,e)	
(a,d)	b - h - f - e	
a - e - d	(b,f)	
(a,e)	b - h - f	
a - e	....	
....		



```
nx.has_path(G, 'a', 'c')            # False
nx.has_path(G, 'a', 'b')            # True
nx.shortest_path(G, 'a', 'b')       # ['a','e','f',h','b']
nx.shortest_path_length(G,'a','b') # 4
nx.shortest_path(G, 'a')            # dictionary
nx.shortest_path_length(G, 'a')     # dictionary
nx.shortest_path(G)                 # all pairs
nx.shortest_path_length(G)          # all pairs
nx.average_shortest_path_length(G) # error
G.remove_node('c')                  # make G connected
nx.average_shortest_path_length(G) # now okay
```

```
{'a': {'a': ['a'],
  'b': ['a', 'e', 'f', 'h', 'b'],
  'd': ['a', 'e', 'd'],
  'e': ['a', 'e'],
  'f': ['a', 'e', 'f'],
  'g': ['a', 'e', 'f', 'g'],
  'h': ['a', 'e', 'f', 'h'],
  'i': ['a', 'e', 'f', 'h', 'i']},
 'b': {'a': ['b', 'h', 'f', 'e', 'a'],
```



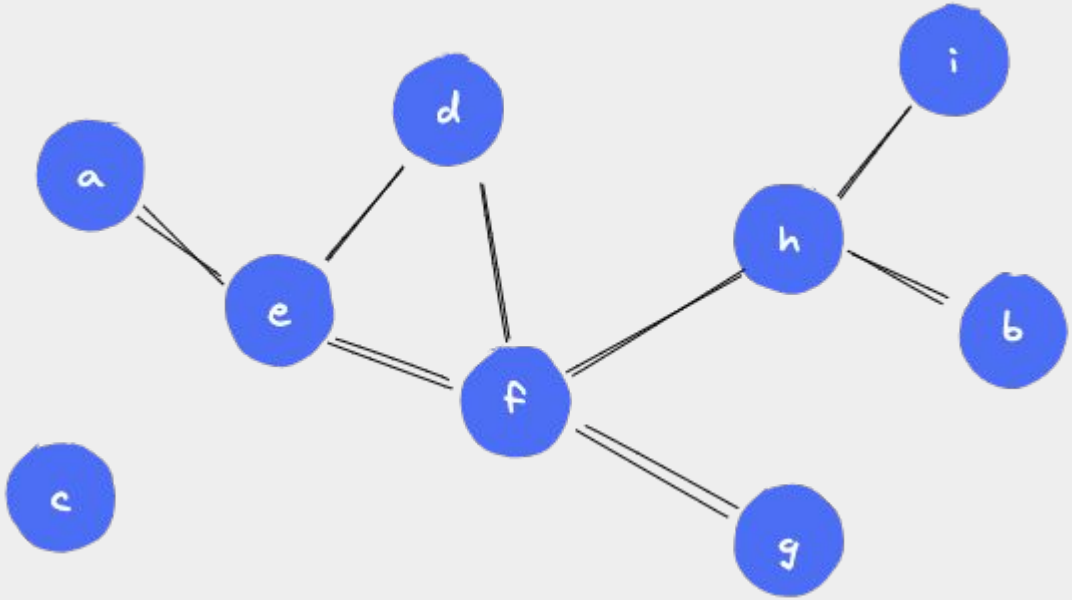

```
{'a': ['a'],
 'b': ['a', 'e', 'f', 'h', 'b'],
 'd': ['a', 'e', 'd'],
 'e': ['a', 'e'],
 'f': ['a', 'e', 'f'],
 'g': ['a', 'e', 'f', 'g'],
 'h': ['a', 'e', 'f', 'h'],
 'i': ['a', 'e', 'f', 'h', 'i']}
```

```
{'a': 0, 'b': 4, 'd': 2, 'e': 1, 'f': 2, 'g': 3, 'h': 3, 'i': 4}
```

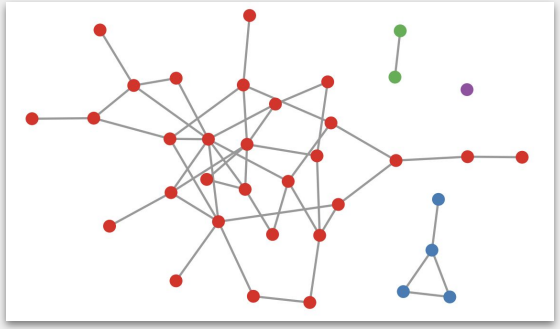
NetworkX
Network Analysis in Python

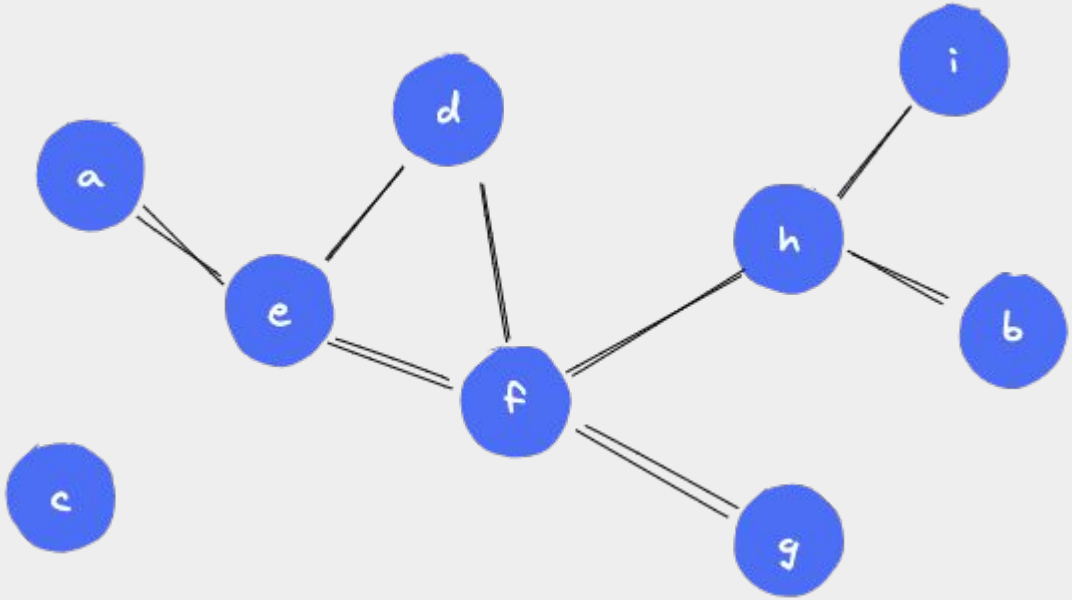
Connected Components

If two nodes cannot be connected by a walk, then they are on different **connected components** . Connected components are subgraphs whose nodes can be reached from one another by following the edges of the network.



Giant Connected Components (GCC)





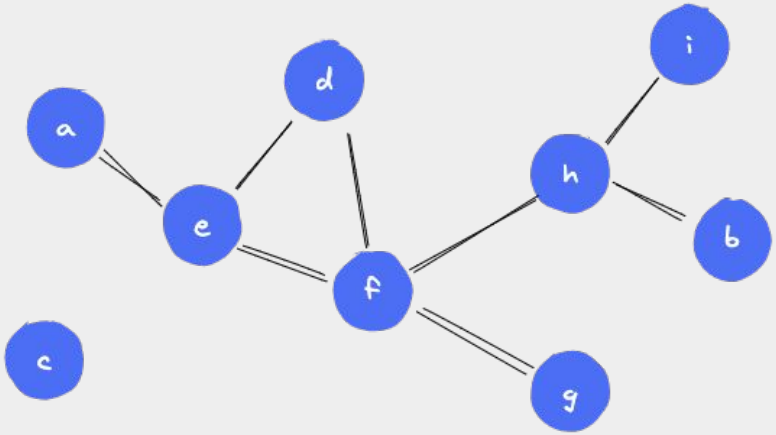
```
# G is connected or not?
nx.is_connected(G)
False

# interact under all connected component of G
for component in nx.connected_components(G):
        print(component)
{'a', 'i', 'e', 'g', 'h', 'd', 'b', 'f'}
{'c'}

# how many connected components has G?
nx.number_connected_components(G)
2

# which connected component is a node N?
nx.node_connected_component(G,"a")
{'a', 'b', 'd', 'e', 'f', 'g', 'h', 'i'}
```

How these function are implemented?



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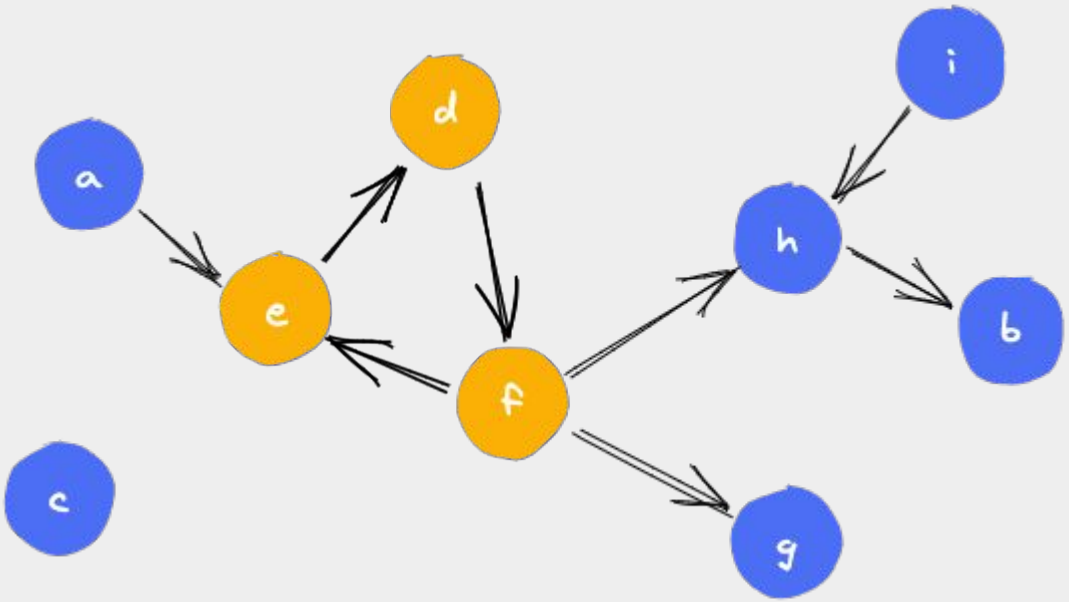
Strongly & Weakly Components (SCC vs WCC)

What does it mean a strongly or weakly network?



NetworkX
Network Analysis in Python

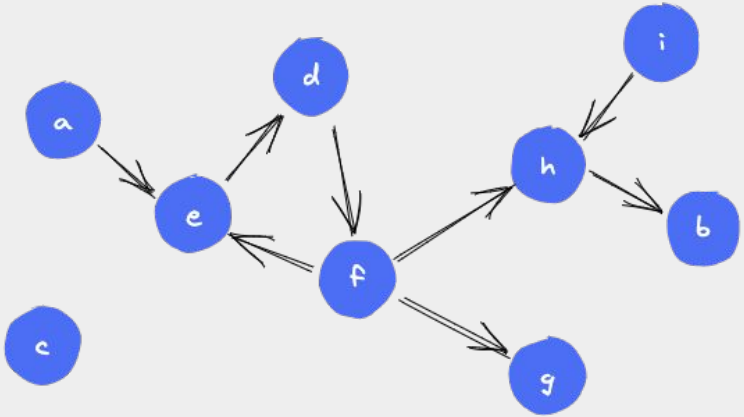
```
nx.is_strongly_connected(G)
False

nx.is_weakly_connected(G)
False

list(nx.weakly_connected_components(G))
[{'a', 'b', 'd', 'e', 'f', 'g', 'h', 'i'}, {'c'}]

list(nx.strongly_connected_components(G))
[{'b'}, {'h'}, {'g'}, {'d', 'e', 'f'}, {'a'}, {'i'}, {'c'}]

nx.number_strongly_connected_components(G)
7
```



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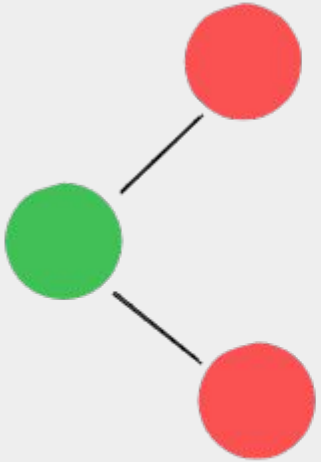
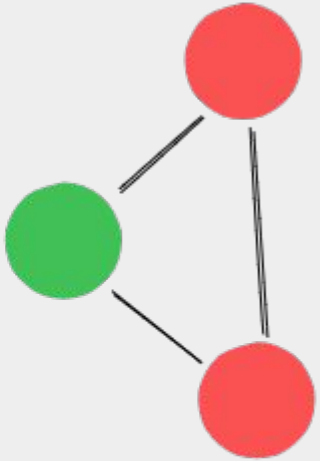
Network Analysis in Python

Friend of a Friend



Triad



Triangle

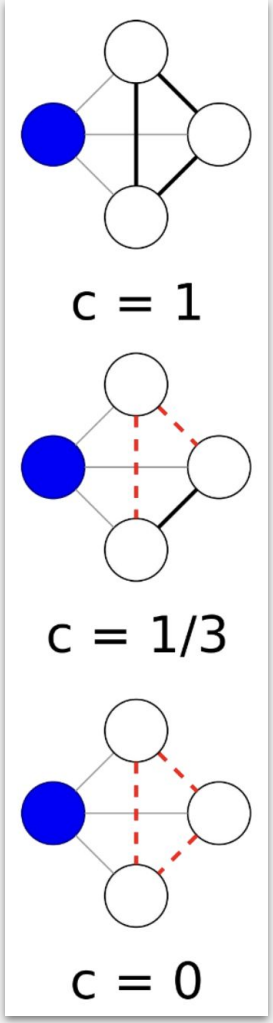
Clustering Coefficient

Some social theories consider triads essential units of social network analysis.

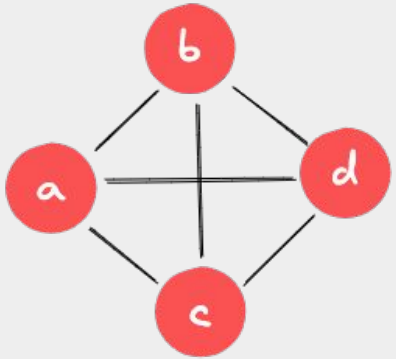
- The **clustering coefficient** is the fraction of possible triangles that contain the ego
- Think of the clustering coefficient as a measure of the **"stardom"**

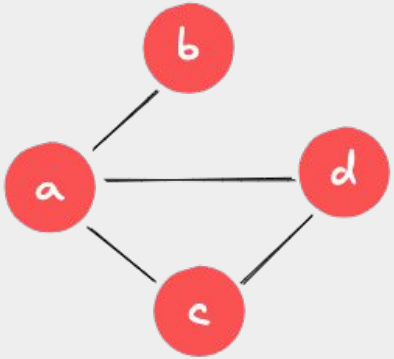
$$C(i) = \frac{\tau(i)}{\tau_{max}(i)} = \frac{\tau(i)}{\binom{k_i}{2}} = \frac{2\tau(i)}{k_i(k_i - 1)}$$

$$C = \frac{\sum_{i;k_i>1} C(i)}{N_{k>1}}$$

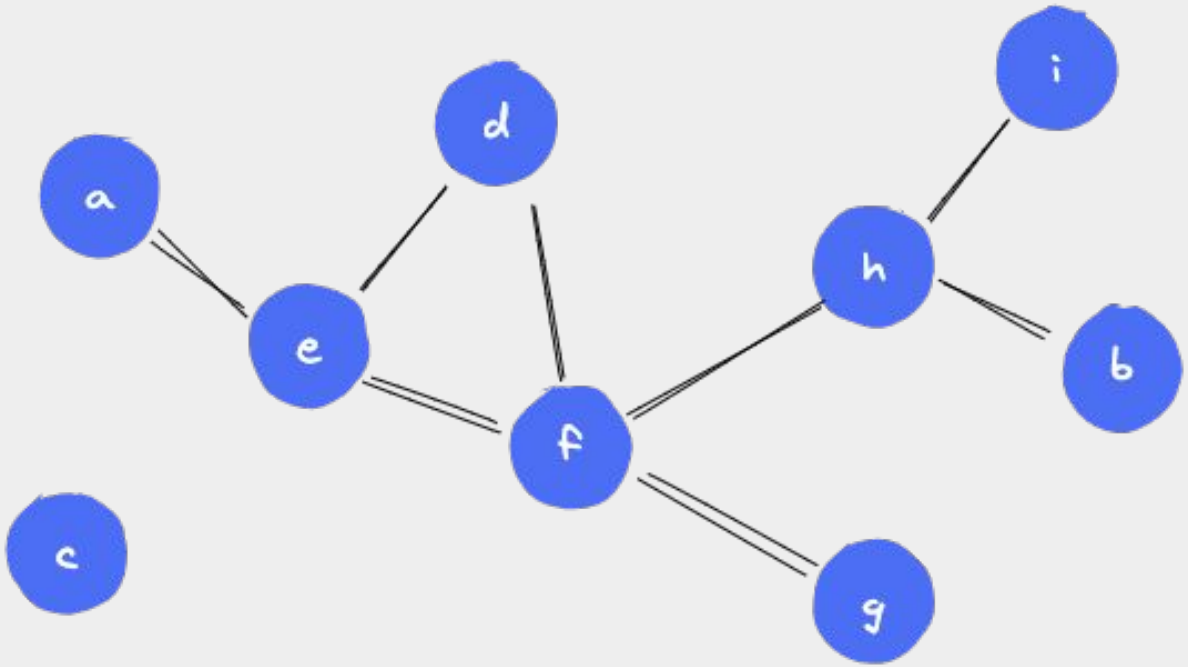


```
nx.triangles(G)
{'a': 3, 'b': 3, 'c': 3, 'd': 3}

nx.clustering(G,"a")
1.0

nx.clustering(G)
{'a': 1.0, 'b': 1.0, 'c': 1.0, 'd': 1.0}

nx.average_clustering(G)
1

G.remove_edge("b","c")
G.remove_edge("b","d")

nx.clustering(G)
{'a': 0.3333333333333333, 'b': 0, 'c': 1.0, 'd': 1.0}

nx.average_clustering(G)
0.5833333333333333
```

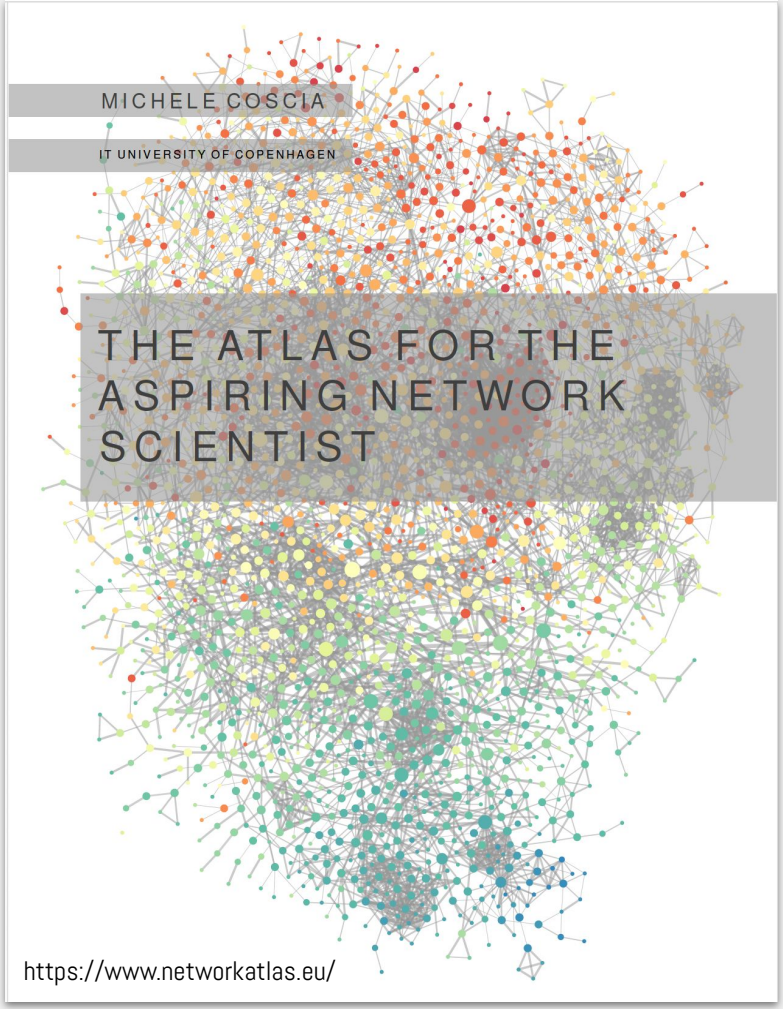




NetworkX
Network Analysis in Python




```
nx.average_clustering(G)
0.16666666666666666
```



Further Reading

Chapter 10 Paths & Walks
Chapter 12 Density
Chapter 13 Shortest Paths